

Practical 15 - Multinomial and Ordinal Logistic Regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `multinomial.csv`

Introduction

In this practical we consider data from the 2003-2004 US National Health and Nutrition Examinations Survey (NHANES), fitting models that relate a categorised version of systolic blood pressure (SBP) to age and sex. We will categorise blood pressure according to SBP as follows.

Category	SBP
1. Normal	< 120
2. Prehypertension	$120 - 139$
3. High (stage 1)	$140 - 159$
4. High (stage 2)	≥ 160

Given the above classification system, it is of interest to model how the probability of falling into the four categories depends on age and sex.

Data

The dataset is called `multinomial.csv` and contains three variables as below.

Variable	Description
sex	“Female”, “Male”
ageyrs	Age in (whole number of) years
sbp	Systolic blood pressure (mmHg)

Aims

- To understand how to interpret coefficients from multinomial and ordinal logistic regression models.
- To understand how to calculate estimated probabilities from multinomial and ordinal logistic regression models.
- To understand how to compare the fits of multinomial and ordinal logistic regression models.

Analysis

Question 1

Load the data into R. Then create a new categorical SBP variable **sbpcat**, categorised as above. One way to do this in R is as follows.

```
multi$sbpcat <- cut(multi$sbp,
                    breaks = c(0, 120, 140, 160, 300),
                    labels = c("Normal", "Prehypertension", "High1", "High2"),
                    right = FALSE,
                    ordered = T) # make it ordered
```

Tabulate **sbpcat** to see its distribution.

Also, to aid interpretation of the intercept estimates in our models, centre the age variable at 50 years.

```
multi$cen_age <- multi$ageyrs - 50
```

SOLUTIONS

```
multi <- read.csv("multinomial.csv")

multi$sbpcat <- cut(multi$sbp,
  breaks = c(0, 120, 140, 160, 300),
  labels = c("Normal", "Prehypertension", "High1", "High2"),
  right = FALSE,
  ordered = T) # make it ordered

multi$cen_age <- multi$ageyrs - 50

table(multi$sbpcat)
```

Normal	Prehypertension	High1	High2
1220	896	287	145

Most participants are in the 'Normal' category, with the numbers in each successive category becoming progressively smaller.

Question 2

Use the `multinom` function in the `nnet` package to fit a multinomial logistic model to the categorised SBP variable with sex and centred age as covariates.

Carefully describe the interpretation of each of the estimated parameters.

SOLUTIONS

```
library(nnet)
multi_sexage <- multinom(sbpcat ~ sex + cen_age, data = multi)
```

```
# weights: 16 (9 variable)
initial value 3532.278032
iter 10 value 2600.578677
final value 2511.160504
converged
```

```
summary(multi_sexage)
```

```
Call:
multinom(formula = sbpcat ~ sex + cen_age, data = multi)
```

Coefficients:

	(Intercept)	sexMale	cen_age
Prehypertension	-0.3877375	0.6070769	0.04221010
High1	-1.5405251	0.3098489	0.07866921
High2	-2.3134874	-0.3101419	0.10677553

Std. Errors:

	(Intercept)	sexMale	cen_age
Prehypertension	0.07347744	0.09450747	0.002917732
High1	0.11170471	0.14414353	0.004411192
High2	0.15822124	0.19386082	0.006556140

Residual Deviance: 5022.321

AIC: 5040.321

-0.3877347 is the estimated log odds of being in the 'Prehypertension' category in females aged 50, ignoring females in the 'High (stage 1)' and 'High (stage 2)' categories.

0.6070803 is the estimated log odds ratio for having systolic blood pressure in the 'Prehypertension' category comparing males with females, conditional on age, and ignoring people in the 'High (stage 1)' and 'High (stage 2)' categories.

0.0422103 is the estimated log odds ratio for having systolic blood pressure in the 'Prehypertension' category for a one-year increase in age, conditional on sex, and ignoring people in the 'High (stage 1)' and 'High (stage 2)' categories.

The interpretation of the three 'High (stage 1)' parameter estimates are analogous to those given above. Here, however, all of the comparisons are between the 'High (stage 1)' and 'Normal' categories, ignoring people in the 'Prehypertension' and 'High (stage 2)' categories.

The interpretation of the three 'High (stage 2)' parameter estimates are also analogous to those given above. Here, however, all of the comparisons are between the 'High (stage 2)' and 'Normal' categories, ignoring people in the 'Prehypertension' and 'High (stage 1)' categories.

Question 3

From your fitted model, calculate (by hand) the estimated probability of having SBP in the second highest category (`sbpcat = 3`) for a female aged 70 years.

SOLUTIONS

The probability that a female aged 70 (i.e. with centred age = 20) has systolic blood pressure in the *High (stage 1)* category is (using the formulae in the lecture notes)

$$\frac{e^{-1.540525+20 \times 0.078669}}{1 + e^{-.3877347+20 \times 0.042210} + e^{-1.540525+20 \times 0.078669} + e^{-2.313477+20 \times 0.106776}} = 0.232$$

Note that the calculation requires the use of **all three linear predictors**, not just the linear predictor reported for the *High (stage 1)* category.

Question 4

Use the `fitted.values` stored in your fitted model to plot the probabilities against age, first in females. Then create the same plot but for the males.

Check that the predicted probability of a 70 year old female having SBP in the second highest category (`spbcats = 3`) matches that calculated above.

SOLUTIONS

We first create the plot using base R plotting functions:

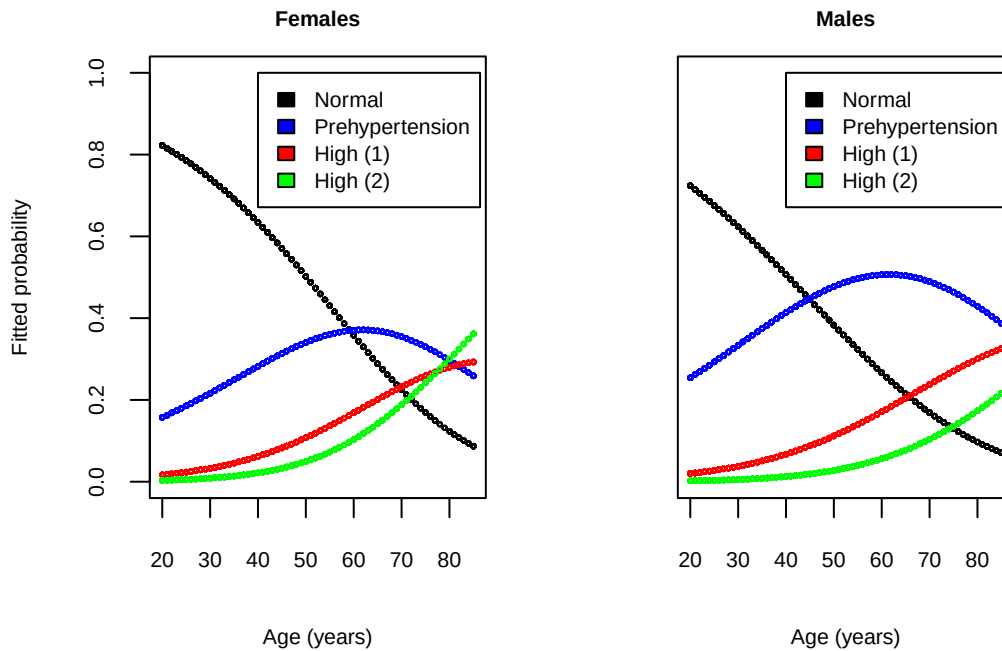
```
#tell it we want a final plot with two columns
#(i.e. two plots side by side)
par(mfrow = c(1, 2))
#reduce space between the two plots
par(mar = c(4, 4, 2, 1))
plot(multi$age[multi$sex=="Female"],
     multi_sexage$fitted.values[multi$sex=="Female",1],
     xlab="Age (years)",
     ylab="Fitted probability",
     main="Females",
     ylim=c(0,1),
     cex=0.3,
     cex.main=0.7,
     cex.lab=0.7,
     cex.axis=0.7)
points(multi$age[multi$sex=="Female"],
       multi_sexage$fitted.values[multi$sex=="Female",2],
       col="blue", cex=0.3)
points(multi$age[multi$sex=="Female"],
       multi_sexage$fitted.values[multi$sex=="Female",3],
       col="red", cex=0.3)
```

```

points(multi$age[multi$sex=="Female"],
      multi_sexage$fitted.values[multi$sex=="Female",4],
      col="green", cex=0.3)
legend(40,1, legend=c("Normal", "Prehypertension", "High (1)", "High (2)"),
      fill=c("black","blue","red","green"), cex=0.7)

plot(multi$age[multi$sex=="Male"],
      multi_sexage$fitted.values[multi$sex=="Male",1], xlab="Age (years)",
      main="Males",
      ylim=c(0,1),
      ylab="",
      yaxt = "n",
      cex=0.3,
      cex.main=0.7,
      cex.lab=0.7,
      cex.axis=0.7)
points(multi$age[multi$sex=="Male"],
      multi_sexage$fitted.values[multi$sex=="Male",2],
      col="blue", cex=0.3)
points(multi$age[multi$sex=="Male"],
      multi_sexage$fitted.values[multi$sex=="Male",3],
      col="red", cex=0.3)
points(multi$age[multi$sex=="Male"],
      multi_sexage$fitted.values[multi$sex=="Male",4],
      col="green", cex=0.3)
legend(40,1,
      legend=c("Normal", "Prehypertension", "High (1)", "High (2)"),
      fill=c("black","blue","red","green"), cex=0.7)

```



```
#reset graphical parameter to usual
par(mfrow = c(1, 1))
```

Alternatively, using `ggplot2`:

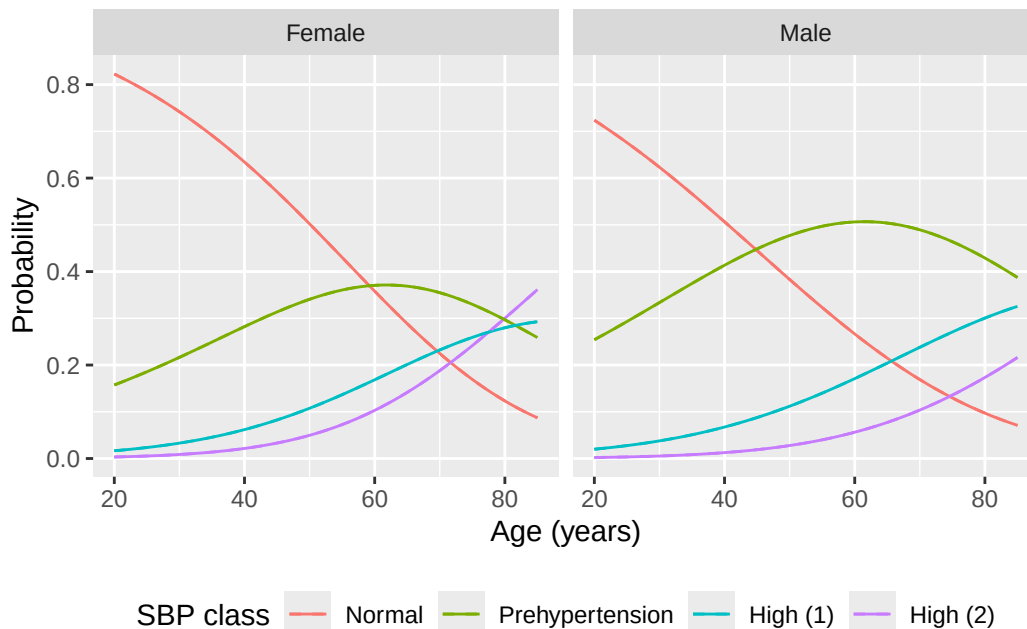
```
# get predictions: note a probability is given for each class, and each row adds up to 1
multinom_pred <- predict(multi_sexage, type="probs")

# add information on sex and age from original dataset
multinom_pred <- cbind(multinom_pred, multi[,c('sex', 'ageyrs')])

# transform to long format to facilitate plotting
library(tidyverse)
multinom_pred_long <- multinom_pred %>%
  mutate(id = 1:length(sex)) %>% # create ID variable
  pivot_longer(Normal:High2, names_to="SBP_class", values_to= "Probability")

#convert SBP class to factor so order is correct
multinom_pred_long <- multinom_pred_long %>%
  mutate(SBP_class = factor(SBP_class,
    levels = c("Normal", "Prehypertension", "High1", "High2"),
    labels = c("Normal", "Prehypertension", "High (1)", "High (2)")))
```

```
# plot, for each SBP_class, the predicted probabilities against age, separately for
# each sex
multinom_pred_long %>%
  ggplot(aes(ageyrs, Probability, colour = SBP_class)) + # plot
  geom_line() +
  theme(legend.position = "bottom") +
  facet_wrap(~sex) +
  labs(x="Age (years)", colour="SBP class") +
  geom_line(data = multinom_pred_long, linetype=2)
```



Notice that the way in which the fitted probabilities of being in each category change with increasing age is not always monotonic. The multinomial logistic regression model does not preclude non-monotonic dependencies on covariates.

```
summary(multi_sexage$fitted.values[multi$sex=="Female" & multi$cen_age==20,3])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2323	0.2323	0.2323	0.2323	0.2323	0.2323

The predicted probability that a female aged 70 (i.e., with centred age = 20) has systolic blood pressure that is in the 'High (stage 1)' category matches the calculation in question 3.

Question 5

Now use the `polr` function in the `MASS` package to fit an ordinal logistic regression model to the categorised SBP variable with sex and centred age as covariates.

Carefully describe the interpretation of each of the estimated parameters.

SOLUTIONS

```
library(MASS)
polr_sexage <- polr(sbpcat ~ sex + cen_age, data=multi)
summary(polr_sexage)
```

Call:

```
polr(formula = sbpcat ~ sex + cen_age, data = multi)
```

Coefficients:

	Value	Std. Error	t value
sexMale	0.27952	0.079594	3.512
cen_age	0.05744	0.002395	23.982

Intercepts:

	Value	Std. Error	t value
Normal Prehypertension	-0.1949	0.0634	-3.0732
Prehypertension High1	1.8744	0.0747	25.0906
High1 High2	3.2483	0.1019	31.8719

Residual Deviance: 5078.576

AIC: 5088.576

The estimate 0.0574355 is the estimated (common) log odds ratio for a one-year increase in age conditional on sex. Under the ordinal logistic model this odds ratio is simultaneously

- i) the log odds ratio for `sbpcat` equalling 2 or more rather than 1,
- ii) the log odds ratio for `sbpcat` equalling 3 or more rather than 2 or less, and
- iii) the log odds ratio for `sbpcat` equalling 4 rather than 3 or less.

All three of these log odds ratios are assumed to be equal (the proportional odds assumption).

From the course manual, we know that the estimated cut-points are the estimates of minus the intercept parameters. Thus, +0.1949374 is the estimated log odds of `sbpcat` equalling 2 or more for a female aged 50 (the age of someone for whom the centred age variable is zero). And 1.87444 is the estimated log odds of `sbpcat` equalling 3 or more for a female aged 50, whilst -3.24833 is the estimated log odds of `sbpcat` equalling 4 for a female aged 50.

Question 6

From your fitted ordinal logistic regression model, calculate (by hand) the estimated probability of having SBP in the second highest category (`spbcat = 3`) for a female aged 70 years.

SOLUTIONS

Using the parameter estimates from the ordinal logistic regression model, we can estimate the probability that a female aged 70 (i.e. with centred age = 20) has systolic blood pressure in the *High (stage 1)* category, **or a higher category**, as follows:

$$\frac{\exp(-1.87444 + 20 \times 0.0574355)}{1 + \exp(-1.87444 + 20 \times 0.0574355)} = 0.326$$

We can also estimate the probability that a female aged 70 has systolic blood pressure in the *High (stage 2)* category, or higher, as follows:

$$\frac{\exp(-3.24833 + 20 \times 0.0574355)}{1 + \exp(-3.24833 + 20 \times 0.0574355)} = 0.109$$

Therefore, we can estimate the probability that a female aged 70 has systolic blood pressure in the *High (stage 1)* category by subtraction: $0.326 - 0.109 = 0.217$.

The predicted probability here (0.217) is not very different from that calculated in question 3 for the multinomial logistic regression model (0.232).

Question 7

As for the multinomial model, use the `fitted.values` in your fitted ordinal logistic model object to plot the probabilities of being in each of the four SBP categories against age in females. Then repeat the plot but for the males.

Again check that the predicted probability of a 70 year old female having SBP in the second highest category (`spbcat = 3`) matches that calculated above.

Are the predictions from the ordinal and multinomial logistic regression models markedly different?

SOLUTIONS

We first check our manual calculation:

```
summary(polr_sexage$fitted.values[multi$sex=="Female" & multi$cen_age==20,3])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.217	0.217	0.217	0.217	0.217	0.217

which matches our manual calculation.

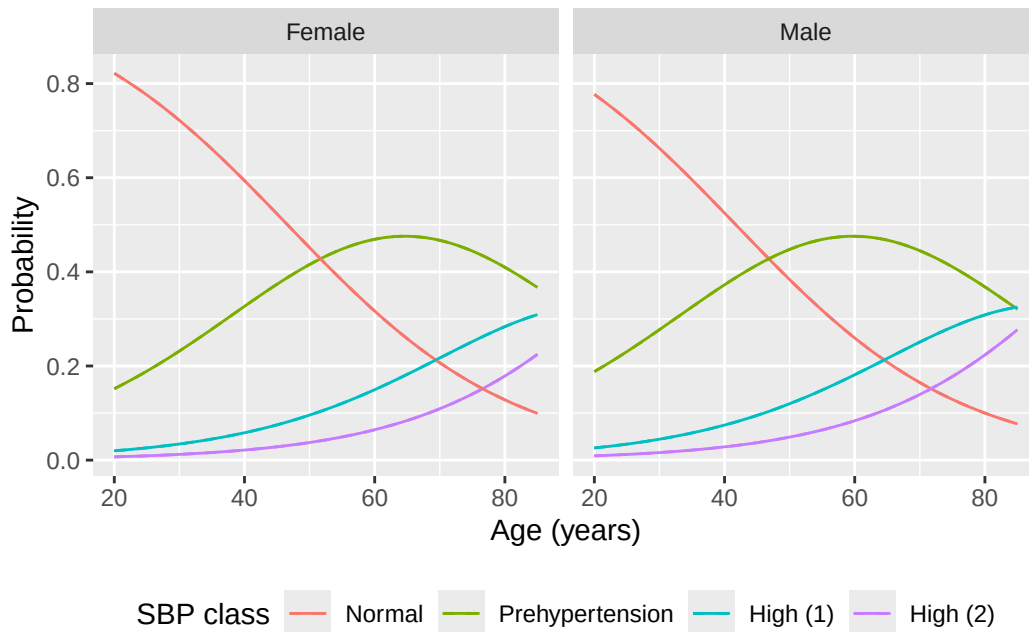
```
# get predictions: note a probability is given for each class, and each row adds up to 1
polr_pred <- predict(polr_sexage, type="probs")

# add information on sex and age from original dataset
polr_pred <- cbind(polr_pred, multi[,c('sex', 'ageyrs')])

# transform to long format to facilitate plotting
library(tidyverse)
polr_pred_long <- polr_pred %>%
  mutate(id = 1:length(sex)) %>% # create ID variable
  pivot_longer(Normal:High2, names_to="SBP_class", values_to= "Probability")

#convert SBP class to factor so order is correct
polr_pred_long <- polr_pred_long %>%
  mutate(SBP_class = factor(SBP_class,
    levels = c("Normal", "Prehypertension", "High1", "High2"),
    labels = c("Normal", "Prehypertension", "High (1)", "High (2)")))

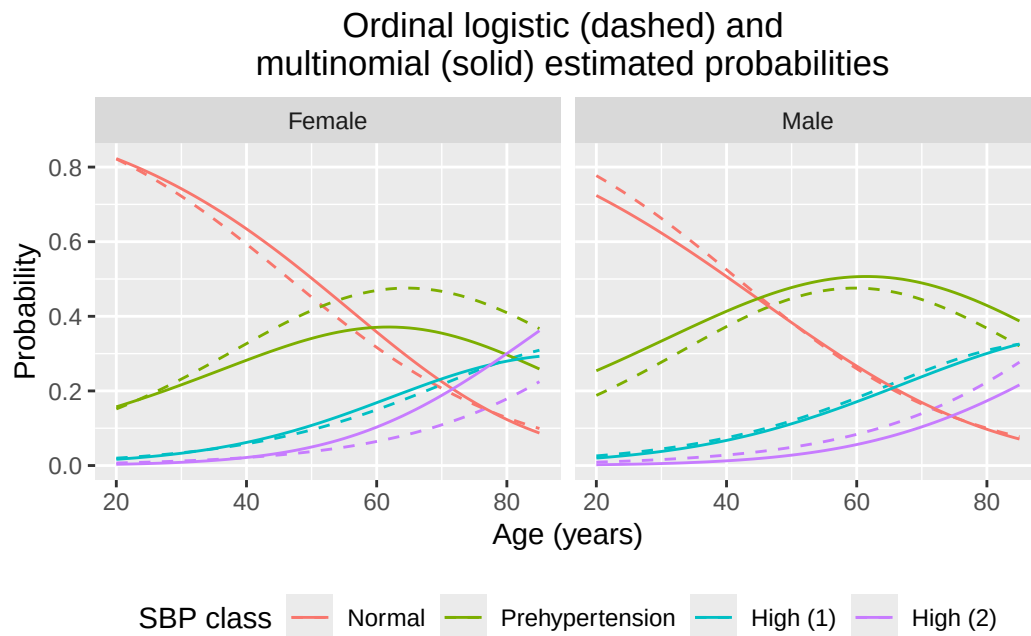
# plot, for each SBP_class, the predicted probabilities against age, separately for
# each sex
polr_pred_long %>%
  ggplot(aes(ageyrs, Probability, colour = SBP_class)) + # plot
  geom_line() +
  theme(legend.position = "bottom") +
  facet_wrap(~sex) +
  labs(x="Age (years)", colour="SBP class") +
  geom_line(data = polr_pred_long, linetype=2)
```



As with the multinomial logistic regression model, the way in which the fitted probabilities of being in each category change with increasing age is not always monotonic. With the ordinal logistic regression model the fitted probabilities of being in the highest and lowest categories are always monotonic functions of the linear predictor, but the fitted probabilities of being in the intermediate categories need not be.

Comparing this graph with the analogous plot for the ordinal logistic regression model suggests some differences, these being most obvious for older females.

```
ggplot() +
  geom_line(
    data = polr_pred_long,
    aes(ageyrs, Probability, colour = SBP_class),
    linetype = "dashed"
  ) +
  geom_line(
    data = multinom_pred_long,
    aes(ageyrs, Probability, colour = SBP_class),
    linetype = "solid"
  ) +
  facet_wrap(~ sex) +
  theme(legend.position = "bottom",
        plot.title = element_text(hjust = 0.5)) +
  labs(x = "Age (years)", colour = "SBP class",
       title = "Ordinal logistic (dashed) and \n multinomial (solid) estimated probabilities")
```



Question 8

To further assess whether the proportional odds assumption is reasonable here compare the log-likelihoods and AIC values from the two models.

SOLUTIONS

```
AIC(multi_sexage)
```

```
[1] 5040.321
```

```
AIC(polr_sexage)
```

```
[1] 5088.576
```

The AIC is markedly lower (better) for the multinomial logistic regression model.

Question 9

As a final check of the ordinal logistic regression model fit, compare the coefficients to those from a series of logistic regression models with different cut-offs. This can be done in different way. One is to create three new variables in the data frame for the three different dichotomisations:

1. Normal vs All other categories
2. Normal or Prehypertension vs Hypertension (Stage 1 or 2)
3. All other categories vs Stage 2 Hypertension

Alternatively, you can use the original continuous **sbp** variable together with the **I()** function to specify the different dichotomised outcomes directly in a call to **glm**.

Fit separate standard logistic regression models for each of the three dichotomised outcomes, with sex and centred age as covariates. Compare the coefficients from each model to the equivalent from the ordinal logistic regression model.

Considering all of the above, which model do you prefer for these data?

SOLUTIONS

```
summary(glm(I(sbp >= 120) ~ sex+cen_age, data=multi, family=binomial))
```

Call:

```
glm(formula = I(sbp >= 120) ~ sex + cen_age, family = binomial,  
    data = multi)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.04602	0.06628	0.694	0.487
sexMale	0.49244	0.08873	5.550	2.86e-08 ***
cen_age	0.05425	0.00273	19.872	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 3527.7 on 2547 degrees of freedom
Residual deviance: 3001.5 on 2545 degrees of freedom
AIC: 3007.5

Number of Fisher Scoring iterations: 4

```
summary(glm(I(sbp>= 140) ~ sex+cen_age, data=multi, family=binomial))
```

Call:

```
glm(formula = I(sbp >= 140) ~ sex + cen_age, family = binomial,  
     data = multi)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.670222	0.089142	-18.737	<2e-16 ***
sexMale	-0.227058	0.117310	-1.936	0.0529 .
cen_age	0.063905	0.003496	18.279	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2319.5 on 2547 degrees of freedom
Residual deviance: 1894.7 on 2545 degrees of freedom
AIC: 1900.7

Number of Fisher Scoring iterations: 5

```
summary(glm(I(sbp>= 160) ~ sex+cen_age, data=multi, family=binomial))
```

Call:

```
glm(formula = I(sbp >= 160) ~ sex + cen_age, family = binomial,  
     data = multi)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.983977	0.148548	-20.088	< 2e-16 ***
sexMale	-0.665255	0.183094	-3.633	0.00028 ***
cen_age	0.072095	0.005827	12.373	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1112.82 on 2547 degrees of freedom

Residual deviance: 900.04 on 2545 degrees of freedom
AIC: 906.04

Number of Fisher Scoring iterations: 7

The three estimated log odds ratios for a one unit increase in age are 0.054 (95% CI 0.049 to 0.060), 0.064 (0.057 to 0.071) and 0.072 (0.061 to 0.084) - some variation, but not not hugely different from one another or from the estimate obtained assuming a common log odds ratio (0.057 (0.053 to 0.062)). However, the three estimated log odds ratios for the comparison between males and females vary markedly: 0.49 (0.67 to 0.32), -0.23 (0.00 to -0.46) and -0.66 (-0.31 to -1.02). There is evidence that conditional on age, males are less likely to be in the highest category of blood pressure vs. the lower levels, and also more likely to be in the lowest category of blood pressure vs. the higher levels.

The comparisons we have made in questions 8 and 9 suggest that the multinomial logistic regression model better describes the data here. The predicted probabilities from the two models are not hugely different different, but nonetheless we would probably prefer those from the multinomial logistic regression model.